

Martingales in multi-type general branching processes

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A branching-process point of view

A branching process is a random population in which individuals reproduce independently.

Examples:

- Galton–Watson process: generations, discrete time.
- Bellman–Harris process: random lifetimes, births at death.
- Branching random walk: individuals carry spatial positions.
- Crump–Mode–Jagers process: births occur at arbitrary ages.

The CMJ framework is broad enough to include age-dependent and point-process reproduction.

Genealogical notation

Let

$$\mathcal{U} := \bigcup_{n \geq 0} \mathbb{N}^n$$

be the Ulam–Harris tree.

- The ancestor is $\emptyset$.
- If $u \in \mathcal{U}$, its children are $u1, u2, \dots$
- The birth time of u is denoted by $S(u)$.
- The parent of $u \neq \emptyset$ is $\overleftarrow{u}$.

In the single-type CMJ process, every individual u has an independent copy

$$\xi_u = \sum_{j \geq 1} \delta_{X_j(u)}$$

of a point process ξ on $[0, \infty)$.

Here $X_j(u)$ is the age at which u gives birth to child uj , so

$$S(uj) = S(u) + X_j(u).$$

The reproduction measure and the Malthusian parameter

The mean reproduction measure is

$$\mu(B) := \mathbb{E}\xi(B), \quad B \subseteq [0, \infty).$$

Its Laplace transform is

$$\mathcal{L}\mu(z) := \int_0^\infty e^{-zt} \mu(dt),$$

whenever the integral is well-defined.

We assume that there is a *Malthusian parameter* $\alpha > 0$, that is, the unique real solution of

$$\mathcal{L}\mu(\alpha) = 1.$$

This is the exponential growth rate of the population.

Random characteristics

A random characteristic is a stochastic process

$$\varphi = (\varphi(t))_{t \in \mathbb{R}}$$

which describes the contribution of one individual at age t . Usually $\varphi(t) = 0$ for $t < 0$.

We assume that (φ_u, ξ_u) are i.i.d.

The CMJ process counted with characteristic φ is

$$Z_t^\varphi := \sum_{u \in \mathcal{U}} \varphi_u(t - S(u)).$$

Examples:

$$\varphi_u(t) = \mathbb{1}_{\{t \geq 0\}}$$

counts all individuals born by time t ,

$$\varphi_u(t) = \mathbb{1}_{\{0 \leq t < \zeta_u\}}$$

counts individuals alive at time t ,

$$\varphi_u(t) = f(t) \mathbb{1}_{\{t \geq 0\}}$$

counts an age-weighted population.

The coming generation

For $t \geq 0$, define the coming generation

$$\mathcal{C}_t := \{u \in \mathcal{U} : S(u) > t, S(\overleftarrow{u}) \leq t\}.$$

These are the individuals not yet born at time t , but whose parents have already been born.

The coming generation is the continuous-time analogue of a generation boundary in a branching random walk.

Nerman's martingale

The classical Nerman martingale is

$$W_t := \sum_{u \in \mathcal{C}_t} e^{-\alpha S(u)}$$

with respect to

$$\mathcal{F}_t = \sigma(\xi_u : S(u) \leq t).$$

Why is it a martingale? Since

$$\mathbb{E} \sum_j e^{-\alpha X_j} = \mathcal{L}\mu(\alpha) = 1,$$

the expected future discounted mass descending from a coming-generation individual u is exactly $e^{-\alpha S(u)}$.

Therefore, for $s \geq t$,

$$\mathbb{E}[W_s \mid \mathcal{F}_t] = W_t.$$

Under standard $X \log X$ assumption,

$$W_t \longrightarrow W \quad \text{a.s. and in } L^1.$$

Nerman's law of large numbers

For a broad class of characteristics φ ,

$$e^{-\alpha t} Z_t^\varphi \longrightarrow a_\varphi W \quad \text{a.s.}$$

The constant has the renewal form

$$a_\varphi = \frac{\int_0^\infty e^{-\alpha s} \mathbb{E} \varphi(s) \, ds}{\int_0^\infty s e^{-\alpha s} \mu(ds)},$$

under the relevant non-lattice and integrability assumptions.

Interpretation:

- $e^{\alpha t}$ is deterministic exponential growth,
- W is the random factor that depends on $(\xi_u)_{u \in \mathcal{U}}$,
- a_φ depends only on the characteristic and on the intensity measure μ .

Representing W_t through a characteristic

We want to express

$$W_t = \sum_{v \in \mathcal{C}_t} e^{-\alpha S(v)}$$

as a CMJ-type sum.

Write $v = uj$. Then

$$S(v) = S(u) + X_j(u).$$

The condition $v \in \mathcal{C}_t$ is precisely

$$S(u) \leq t < S(u) + X_j(u).$$

Hence

$$W_t = \sum_u e^{-\alpha S(u)} \sum_j e^{-\alpha X_j(u)} \mathbb{1}_{\{0 \leq t - S(u) < X_j(u)\}}.$$

A weighted coming-generation characteristic

Define

$$\phi_{\alpha,u}(r) := e^{\alpha r} \sum_j e^{-\alpha X_j(u)} \mathbb{1}_{\{0 \leq r < X_j(u)\}}.$$

Then

$$\begin{aligned} e^{-\alpha t} \mathcal{Z}_t^{\phi_\alpha} &= e^{-\alpha t} \sum_u e^{\alpha(t-S(u))} \sum_j e^{-\alpha X_j(u)} \mathbb{1}_{\{0 \leq t-S(u) < X_j(u)\}} \\ &= \sum_u e^{-\alpha S(u)} \sum_j e^{-\alpha X_j(u)} \mathbb{1}_{\{0 \leq t-S(u) < X_j(u)\}} \\ &= W_t. \end{aligned}$$

Thus Nerman's martingale is a CMJ process.

The limit W as a CMJ limit

A telescoping identity over the tree gives

$$W_t = 1 + \sum_u e^{-\alpha S(u)} \left(\sum_j e^{-\alpha X_j(u)} - 1 \right) \mathbb{1}_{\{S(u) \leq t\}}.$$

Letting $t \rightarrow \infty$ and subtracting,

$$W - W_t = \sum_u e^{-\alpha S(u)} \left(\sum_j e^{-\alpha X_j(u)} - 1 \right) \mathbb{1}_{\{S(u) > t\}}.$$

Define

$$\psi_{\alpha,u}(r) := e^{\alpha r} \left(\sum_j e^{-\alpha X_j(u)} - 1 \right) \mathbb{1}_{\{r < 0\}}.$$

Then

$$W - W_t = e^{-\alpha t} \mathcal{Z}_t^{\psi_\alpha}.$$

Consequently,

$$\mathcal{Z}_t^{\phi_\alpha + \psi_\alpha} = e^{\alpha t} W.$$

Other roots of the characteristic equation

Besides α , consider complex roots

$$\lambda \in \mathbb{C}, \quad \mathcal{L}\mu(\lambda) = 1.$$

For such λ , the process

$$W_t(\lambda) := \sum_{u \in \mathcal{C}_t} e^{-\lambda S(u)}$$

is again a complex-valued Nerman–Biggins type martingale.

Under suitable moment assumptions it converges in L^2 whenever

$$\operatorname{Re} \lambda > \frac{\alpha}{2}$$

As before, one can construct characteristics ϕ_λ and ψ_λ such that

$$\mathcal{Z}_t^{\phi_\lambda + \psi_\lambda} = e^{\lambda t} W(\lambda).$$

Multiple roots and matrix exponentials

Assume that λ is a root of multiplicity k_λ of

$$\mathcal{L}\mu(z) = 1.$$

Let N be the nilpotent Jordan block with ones on the superdiagonal and set

$$B_\lambda := \lambda I + N.$$

Define

$$\exp(\lambda, t) := e^{tB_\lambda} = e^{\lambda t} \begin{pmatrix} 1 & t & t^2/2! & \cdots & t^{k_\lambda-1}/(k_\lambda-1)! \\ 0 & 1 & t & \cdots & t^{k_\lambda-2}/(k_\lambda-2)! \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & t \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix}.$$

The martingale identity for a multiple root

Since λ is a root of multiplicity k_λ ,

$$\mathcal{L}\mu(\lambda) = 1, \quad (\mathcal{L}\mu)'(\lambda) = \dots = (\mathcal{L}\mu)^{(k_\lambda-1)}(\lambda) = 0.$$

Hence

$$\mathbb{E} \int \exp(\lambda, -x) \xi(dx) = I_{k_\lambda}.$$

Indeed, the left-hand side is the triangular matrix

$$\begin{pmatrix} \mathcal{L}\mu(\lambda) & -\mathcal{L}\mu'(\lambda) & \mathcal{L}\mu''(\lambda)/2! & \dots \\ 0 & \mathcal{L}\mu(\lambda) & -\mathcal{L}\mu'(\lambda) & \dots \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \mathcal{L}\mu(\lambda) \end{pmatrix} = I_{k_\lambda}$$

Polynomial martingales

The process

$$\mathbf{W}_t(\lambda) := \sum_{v \in \mathcal{C}_t} \exp(\lambda, -S(v))$$

is a matrix-valued martingale.

Its entries are consists of

$$W_t^{(\ell)}(\lambda) = \sum_{v \in \mathcal{C}_t} S(v)^\ell e^{-\lambda S(v)}, \quad \ell = 0, \dots, k_\lambda - 1.$$

Under suitable moment assumptions and $\operatorname{Re} \lambda > \frac{\alpha}{2}$, these martingales converge to a limit $\mathbf{W}(\lambda)$, and one can choose matrix-valued characteristics $\phi_\lambda, \psi_\lambda$ such that

$$\mathcal{Z}_t^{\phi_\lambda + \psi_\lambda} = \exp(\lambda, t) \mathbf{W}(\lambda).$$

Connection with fluctuations

In particular, for $l < k_\lambda$ we can construct a characteristic $\phi_{\lambda,l}$ such that

$$\mathcal{Z}_t^{\phi_{\lambda,l}} = e^{\lambda t} (t^l W^{(0)}(\lambda) + \sum_{j>1} c_j t^{l-j} W^{(j)}(\lambda))$$

$$\mathbb{E} \mathcal{Z}_t^{\phi_{\lambda,l}} = e^{\lambda t} t^l$$

For a general characteristic φ , the mean asymptotics identify the intermediate deterministic shapes.

A typical expansion has the form

$$\mathcal{Z}_t^\varphi = a_\varphi e^{\alpha t} W + H_\varphi(t) + \text{fluctuation term at scale } e^{\alpha t/2},$$

where

$$H_\varphi(t) = \sum_{\lambda: \alpha/2 \leq \operatorname{Re} \lambda < \alpha} \sum_j c_{\lambda,j}(\omega) t^j e^{\lambda t}.$$

The random coefficients are built from the martingale limits $W^{(l)}(\lambda)$.

A classical continuous-time multi-type branching process

Before general multi-type CMJ, consider a Markovian benchmark.

Suppose individuals have types $1, \dots, p$ and reproduce according to exponential/Poisson mechanisms, so that the population vector

$$N(t) = (N^1(t), \dots, N^p(t))$$

has mean semigroup

$$\mathbb{E}_i N(t) = e_i \cdot e^{tA}.$$

Here A is the mean generator matrix.

Spectral decomposition of A gives terms

$$e^{\lambda t} t^j,$$

where polynomial powers come from Jordan blocks of A .

Martingales from the spectral decomposition of A

Let λ be an eigenvalue of A , and let π_λ denote the spectral projection onto the associated generalized eigenspace.

In the Markovian multi-type setting, the corresponding projected process has the martingale form

$$e^{-tA}\pi_\lambda N(t).$$

When

$$\operatorname{Re} \lambda > \frac{\alpha}{2},$$

this martingale has a non-trivial limit $W(\lambda)$.

Then we have expansion of the form

$$N(t) = \sum_{\lambda: \operatorname{Re} \lambda > \alpha/2} e^{tA}\pi_\lambda W(\lambda) + \text{Gaussian fluctuations}.$$

Multi-type CMJ processes

Now let the type space be finite:

$$\mathcal{S} = \{1, \dots, p\}.$$

An individual u has type $\tau(u) \in \mathcal{S}$. An individual of type i produces children of type j according to a point process

ξ^{ij} = birth ages of type j children of a type i parent.

The mean reproduction measures are

$$\mu_{ij}(B) := \mathbb{E}_i \xi^{ij}(B).$$

Collect them into a matrix-valued measure

$$\mu = (\mu_{ij})_{1 \leq i, j \leq p}.$$

The matrix Laplace transform

The matrix Laplace transform is

$$\mathcal{L}\mu(z) := \left(\int_0^\infty e^{-zt} \mu_{ij}(dt) \right)_{i,j=1}^p.$$

The Malthusian parameter α is determined by

$$\rho(\mathcal{L}\mu(\alpha)) = 1,$$

where ρ denotes spectral radius.

The scalar equation

$$\mathcal{L}\mu(z) = 1$$

becomes the matrix characteristic equation

$$\det(I - \mathcal{L}\mu(z)) = 0.$$

Multi-type characteristics

A multi-type characteristic is type-dependent:

$$\varphi_i(t) = \text{contribution at age } t \text{ of an individual of type } i.$$

The counted process is

$$\mathcal{Z}_t^\varphi = \sum_u \varphi_{\tau(u),u}(t - S(u)).$$

Equivalently, if $\mathbf{e}_i$ is the i -th row basis vector and

$$\varphi(t) = (\varphi_1(t), \dots, \varphi_p(t))^T,$$

then

$$\mathcal{Z}_t^\varphi = \sum_u \mathbf{e}_{\tau(u)} \cdot \varphi_u(t - S(u)).$$

Two sources of polynomial factors

The one-type and Markovian multi-type examples look similar, but the polynomial factors seem to have different origins.

Single-type CMJ	Markovian multi-type branching
multiple root of $1 - \mathcal{L}\mu(z)$	Jordan block of A
resolvent pole	spectral projection/Jordan expansion
$t^j e^{\lambda t}$	$t^j e^{\lambda t}$

A general multi-type CMJ theory should unify these two mechanisms.

The unifying object is the resolvent

$$(I - \mathcal{L}\mu(z))^{-1}.$$

The resolvent as the common object

Assume that near λ the matrix-valued function

$$z \mapsto (I - \mathcal{L}\mu(z))^{-1}$$

has a pole of order k :

$$(I - \mathcal{L}\mu(z))^{-1} = \frac{A_k}{(z - \lambda)^k} + \cdots + \frac{A_1}{z - \lambda} + \text{holomorphic part.}$$

The coefficients $A_1, \dots, A_k$ are constrained by the identity

$$(I - \mathcal{L}\mu(z))(I - \mathcal{L}\mu(z))^{-1} = I.$$

We can use the Taylor expansion

$$I - \mathcal{L}\mu(z) = I - \mathcal{L}\mu(\lambda) - \sum_{m \geq 1} (-1)^m \frac{\mathcal{L}\mu^{(m)}(\lambda)}{m!} (z - \lambda)^m.$$

Relations for the principal part coefficients

Comparing the negative powers of $z - \lambda$ gives

$$\begin{cases} (I - \mathcal{L}\mu(\lambda))A_k = 0, \\ \mathcal{L}\mu'(\lambda)A_k + (I - \mathcal{L}\mu(\lambda))A_{k-1} = 0, \\ -\frac{\mathcal{L}\mu''(\lambda)}{2}A_k + \mathcal{L}\mu'(\lambda)A_{k-1} + (I - \mathcal{L}\mu(\lambda))A_{k-2} = 0, \\ \vdots \end{cases}$$

Equivalently,

$$\begin{pmatrix} \mathcal{L}\mu(\lambda) & -\mathcal{L}\mu'(\lambda) & \mathcal{L}\mu''(\lambda)/2! & \cdots \\ 0 & \mathcal{L}\mu(\lambda) & -\mathcal{L}\mu'(\lambda) & \cdots \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \mathcal{L}\mu(\lambda) \end{pmatrix} \begin{pmatrix} A_1 \\ A_2 \\ \vdots \\ A_k \end{pmatrix} = \begin{pmatrix} A_1 \\ A_2 \\ \vdots \\ A_k \end{pmatrix}$$

Block matrix identity

Then the block matrix

$$\int \exp(\lambda, -x) \otimes \mu(dx)$$

contains precisely the matrices

$$\mathcal{L}\mu(\lambda), \quad \mathcal{L}\mu'(\lambda), \quad \dots,$$

with the factorial and sign conventions dictated by $\exp(\lambda, -x)$.

If

$$\vec{A}_\lambda := \sum_{\ell=1}^k \mathbf{f}_\ell \otimes A_\ell,$$

where $\mathbf{f}_\ell$ is the ℓ -th basis vector, then the previous relations can be written compactly as

$$\mathbb{E} \left(\int \exp(\lambda, -x) \otimes \xi(dx) - I_k \otimes I_p \right) \vec{A}_\lambda = 0.$$

The multi-type Nerman–Biggins martingale

The resolvent coefficients determine the correct martingale.

Define

$$\vec{W}_t(\lambda) := \sum_{u \in \mathcal{C}_t} (I_k \otimes e_{\tau(u)}) \cdot (\exp(\lambda, -S(u)) \otimes I_p) \cdot \vec{A}_\lambda.$$

The fixed-point identity from the previous slide implies that

$$(\vec{W}_t(\lambda))_{t \geq 0}$$

is a martingale.

Conceptually:

- $\exp(\lambda, -S(u))$ gives exponential-polynomial time weights,
- $e_{\tau(u)}$ reads off the type of the coming-generation individual,
- $\vec{A}_\lambda$ selects the correct generalized eigendirection of the resolvent.

Theorem: multi-type complex martingales

Let $\alpha > 0$ be the Malthusian parameter, i.e.

$$\rho(\mathcal{L}\mu(\alpha)) = 1.$$

Let λ be a pole of $(I - \mathcal{L}\mu(z))^{-1}$ of order k , and suppose that

$$\operatorname{Re} \lambda > \frac{\alpha}{q} \quad \text{for some } 1 < q \leq 2.$$

Under the corresponding q -moment condition:

$$\mathbb{E} \left\| \int_0^\infty (1 + s^{k-1}) e^{-s \operatorname{Re} \lambda} \xi(ds) \right\|^q < \infty,$$

the martingale $\vec{W}_t(\lambda)$ is bounded in L^q and converges in L^q and a.s. to $\vec{W}(\lambda)$.

Martingale limits as CMJ limits

As in the single-type case, one can construct vector-valued characteristics

$$\vec{\phi}_\lambda, \quad \vec{\psi}_\lambda,$$

supported respectively on the coming-generation crossing and on the future tail.

They satisfy

$$\mathcal{Z}_t^{\vec{\phi}_\lambda + \vec{\psi}_\lambda} = (\exp(\lambda, t) \otimes 1) \vec{W}(\lambda),$$

up to the chosen row/column convention.

Thus the limiting martingales are not only coefficients in an asymptotic expansion. They can themselves be represented through suitable CMJ characteristics.

This closes the circle:

resolvent pole $\Rightarrow$ martingale $\Rightarrow$ CMJ representation $\Rightarrow$ subleading terms.